PINN RESEARCH BLUEPRINT

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Hard vs. Soft Thermodynamic Constraint Enforcement in Physics-Informed Neural Networks for Sorption in Heterogeneous Porous Media

Implementation Blueprint · Mathematical Foundations ·
Unified Statistical Evaluation Harness · Manuscript Architecture

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1 Research Vision, Gap, and Novel Contributions

1.1 The Central Scientific Question

When physical laws are known and must be respected by a neural network prediction, *how* those laws are enforced matters. Soft constraints penalize violations during training but allow them to persist — especially under extrapolation beyond the training distribution. Hard constraints eliminate violations structurally, by construction, but restrict the optimization landscape and may impose computational costs. The fundamental question is:

Does guaranteeing thermodynamic validity by construction outperform penalizing violations, and at what cost in accuracy, calibration, and computational efficiency?

1.2 Why This Has Not Been Done Before

Hard constraint PINNs exist in fluid dynamics (topology optimization, PDE boundary conditions) and structural mechanics. However:

- No study applies hard constraints to equilibrium isotherm sorption models where constraints have a different mathematical character (algebraic inequalities and functional form requirements, not PDE boundary conditions).
- No study systematically compares *each* physics constraint independently in a factorial design, isolating marginal contributions.
- No study evaluates hard vs. soft constraints specifically under *extrapolation* to out-of-distribution coal ranks and pressures, which is the operationally critical scenario.
- No study combines the hard vs. soft comparison with calibrated Bayesian uncertainty quantification, asking whether hard constraints improve not just accuracy but predictive reliability.

1.3 Potential Novel Contributions

Contribution 1: Systematic hard vs. soft constraint comparison for sorption isotherms in geological materials, using a factorial ablation design that isolates and quantifies the marginal benefit of each constraint independently and jointly.

Contribution 2: Three architectural strategies for hard constraint enforcement in sorption PINNs: sigmoid-bounded output activations (bounds), non-negative pressure pathway networks (monotonicity), and gauge function parameterization (Sips/Arrhenius). The Sips gauge strategy is shown to simultaneously enforce bounds, monotonicity, and isotherm functional form as a unified architectural choice.

Contribution 3: Three-tier extrapolation evaluation — interpolation, rank extrapolation, and pressure extrapolation — as a standardized benchmark for physical validity under distribution shift, showing the greatest hard constraint advantage (to be tested!) precisely where soft constraints fail most.

Contribution 4: Physical violation rate as a first-class evaluation metric with bootstrap confidence intervals, enabling statistical comparison of constraint enforcement strategies on safety-relevant properties, not just accuracy.

Contribution 5: Constraint interaction analysis via 2×2 ANOVA revealing whether hard enforcement of multiple constraints produces superadditive, additive, or antagonistic effects. Ablation design is structured to avoid confounding of interaction terms.

Contribution 6: Physics parameter interpretability from the gauge function Sips variant — the network predicts physically meaningful $[\hat{q}_{\max}, \hat{K}, \hat{n}]$ across coal ranks, validated against classical isotherm literature values and SHAP/ALE attribution. Similarly, the Arrhenius variant yields interpretable $\Delta\hat{H}$ values validatable against calorimetric measurements.

2 Mathematical Foundations

2.1 Soft Constraint Formulation (Baseline)

Theory: Soft constraints enter as additive penalty terms in the loss function. They are differentiable, computationally cheap, and allow gradient-based optimization to trade off constraint satisfaction against data fidelity. The penalty weight λ_p controls this trade-off but must be tuned and can never guarantee zero violations.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_p \mathcal{L}_{\text{physics}} \tag{1}$$

$$\mathcal{L}_{\text{data}} = \mathbb{E}\left[\frac{(q-\hat{q})^2}{2\hat{\sigma}^2} + \tfrac{1}{2}\log\hat{\sigma}^2\right] \quad \text{(heteroscedastic NLL)} \tag{2}$$

$$\mathcal{L}_{\text{physics}} = \mathcal{L}_{\text{Sips}} + \mathcal{L}_{\text{bounds}} + \mathcal{L}_{\text{mono}} + \mathcal{L}_{\text{vH}} \tag{3}$$

$$\mathcal{L}_{\text{Sips}} = \text{MSE}\left(\hat{q},\ \frac{\hat{q}_{\max}(\hat{K}P)^{\hat{n}}}{1+(\hat{K}P)^{\hat{n}}}\right) \tag{4}$$

$$\mathcal{L}_{\text{bounds}} = \text{ReLU}(-\hat{q}) + \text{ReLU}(\hat{q} - \hat{q}_{\max}) \tag{5}$$

$$\mathcal{L}_{\text{mono}} = \frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \text{ReLU}\left(-\frac{\partial\hat{q}}{\partial P}\bigg|_T\right) \tag{6}$$

$$\mathcal{L}_{\text{vH}} = \text{MSE}\left(\frac{\partial \ln\hat{K}}{\partial(1/T)},\ -\frac{\Delta\hat{H}}{R}\right) \tag{7}$$

Fundamental limitation: $\lambda_p \to \infty$ theoretically eliminates violations but destabilizes optimization. In practice, violations persist at any finite λ_p.

2.2 Hard Constraint Strategy 1: Constrained Output Activations (Bounds)

v3 correction: The v2 formulation used $\sigma(z_s) \cdot \text{softplus}(z_q) \cdot \hat{q}_{\max}$, which does not guarantee an upper bound because $\text{softplus}(z_q)$ is unbounded. The formulation below is corrected: a single sigmoid output guarantees $\hat{q} \in (0, \hat{q}_{\max})$ with no additional factors needed.

Theory: Non-negativity and saturation bounds are enforced by replacing the unbounded prediction head with a single sigmoid activation that maps $\mathbb{R} \to (0,1)$, scaled by a positive $\hat{q}_{\max}$. Since $\sigma(z) \in (0,1)$ for all $z \in \mathbb{R}$, the product $\sigma(z_q) \cdot \hat{q}_{\max}$ is strictly in $(0, \hat{q}_{\max})$ by construction, for any finite network output z_q.

$$\hat{q}_{\max} = \text{softplus}(\text{NN}_{q_{\max}}(\mathbf{x})) > 0 \tag{8}$$

$$\hat{q} = \hat{q}_{\max} \cdot \sigma(z_q) \ \in\ (0,\ \hat{q}_{\max}) \tag{9}$$

where z_q is a raw scalar network output and σ is the sigmoid function. This is a simpler, safer formulation than any two-factor product: a single bounded factor is sufficient to enforce both bounds simultaneously.

Cost: Negligible. One sigmoid activation per sample. $\mathcal{L}_{\text{bounds}}$ is dropped from the loss.

2.3 Hard Constraint Strategy 2: Non-Negative Pressure Pathway (Monotonicity)

v3 correction: The v2 formulation used a Partially Input-Convex Neural Network (PICNN), claiming that convexity in P implies $\partial\hat{q}/\partial P \geq 0$. This is incorrect: convexity implies the derivative is *non-decreasing*, not necessarily non-negative. A convex function can still decrease. The formulation below replaces PICNN with a non-negative pressure pathway architecture, which directly and provably guarantees $\partial\hat{q}/\partial P \geq 0$. Additionally, Swish activations on the pressure pathway are replaced by softplus (monotone everywhere); Swish has a non-monotone region near $x \approx -1.3$ that can break the monotonicity guarantee.

Theory: A sufficient condition for $\partial\hat{q}/\partial P \geq 0$ is that every pathway through which P contributes to $\hat{q}$ uses (a) non-negative weights and (b) monotone increasing activations. Under these two conditions, increasing P cannot decrease $\hat{q}$, regardless of the network depth.

For a single pressure-pathway layer:

$$\hat{q}(P, \mathbf{x}_c) = a(\mathbf{x}_c) + \text{softplus}(w \cdot P + g(\mathbf{x}_c)) \tag{10}$$

$$\frac{\partial \hat{q}}{\partial P} = w \cdot \sigma(wP + g(\mathbf{x}_c)) \geq 0 \quad \text{iff } w \geq 0 \tag{11}$$

Since $\sigma(\cdot) > 0$ always, the sign of $\partial\hat{q}/\partial P$ is determined entirely by the sign of w. Non-negativity is enforced via reparameterization during every forward pass:

$$w = \text{softplus}(w_{\text{raw}}) \quad (w_{\text{raw}} \in \mathbb{R},\ w > 0) \tag{12}$$

For a multi-layer pressure pathway with L layers:

$$z_P^{(0)} = P \tag{13}$$

$$z_P^{(\ell+1)} = \text{softplus}\left(\underbrace{W_P^{(\ell)}}_{\geq 0} z_P^{(\ell)} + W_c^{(\ell)} \mathbf{x}_c + b^{(\ell)} \right) \tag{14}$$

$$\hat{q} = W_{\text{out}} z_P^{(L)} + a(\mathbf{x}_c) \tag{15}$$

where $W_P^{(\ell)} = \text{softplus}(W_{P,\text{raw}}^{(\ell)}) \geq 0$ is recomputed at every forward pass.

Critical implementation requirement: Pressure P must enter the network *exclusively* through this constrained pathway. If P is also passed through any standard (unconstrained) layer, the guarantee is lost. The compositional features $\mathbf{x}_c$ may enter freely through unconstrained pathways.

Activation choice: softplus or sigmoid only on the pressure pathway. Swish is not permitted on this pathway because it is non-monotone for $x \lesssim -1.3$, which would break the monotonicity guarantee.

Cost: Moderate. Softplus reparameterization added per layer; one additional backward pass for Jacobian verification. $\mathcal{L}_{\text{mono}}$ is dropped.

2.4 Hard Constraint Strategy 3: Gauge Function Parameterization (Sips)

v3 note: The Sips gauge variant simultaneously enforces bounds, monotonicity, and Sips functional form, provided $\hat{q}_{\max}, \hat{K}, \hat{n} > 0$ and raw physical pressure $P > 0$ is used. This makes separate hard-bounds (Strategy 1) and hard-monotonicity (Strategy 2) components redundant within this variant. The ablation design accounts for this explicitly.

Theory: The Sips isotherm shape is enforced by structuring the network to predict isotherm *parameters* from coal composition, then computing the isotherm analytically. The functional form becomes the output layer itself.

$$\hat{q}(\mathbf{x}_c, P) = \frac{\hat{q}_{\max}(\mathbf{x}_c) \cdot [\hat{K}(\mathbf{x}_c) \cdot P]^{\hat{n}(\mathbf{x}_c)}}{1 + [\hat{K}(\mathbf{x}_c) \cdot P]^{\hat{n}(\mathbf{x}_c)}} \tag{16}$$

$$\hat{q}_{\max} = \text{softplus}(\text{NN}_1(\mathbf{x}_c)) > 0 \tag{17}$$

$$\hat{K} = \text{softplus}(\text{NN}_2(\mathbf{x}_c)) > 0 \tag{18}$$

$$\hat{n} = \text{softplus}(\text{NN}_3(\mathbf{x}_c)) > 0 \tag{19}$$

Structural guarantees (given $P > 0$):

- $\hat{q} > 0$ (positivity): numerator always positive
- $\hat{q} < \hat{q}_{\max}$ (saturation bound): Sips ratio < 1
- $\partial\hat{q}/\partial P \geq 0$ (monotonicity): Sips is monotone increasing in P for all positive parameters

The Sips gauge variant therefore subsumes hard-bounds and hard-monotonicity as a

special case. Combining it with separate bounds or monotonicity enforcement (V7 in v2 design) is architecturally redundant.

Implementation requirement: Raw physical pressure $P > 0$ (in MPa) must be used in the Sips layer, not a scaled or log-transformed version. Assert $P > 0$ in the data pipeline. Pressure enters *only* at the output Sips layer, not into the encoder.

Note: The predicted parameters $[\hat{q}_{\max}, \hat{K}, \hat{n}]$ are directly physically interpretable and validatable against classical isotherm literature. $\mathcal{L}_{\text{Sips}}$ is dropped.

2.5 Hard Constraint Strategy 4: Arrhenius Reparameterization (Van't Hoff)

Theory: Van't Hoff consistency requires $K(T)$ to follow Arrhenius kinetics. Rather than penalizing deviations, the network is parameterized to produce $K(T)$ that satisfies it exactly by definition.

$$\hat{K}(T, \mathbf{x}_c) = \hat{K}_0(\mathbf{x}_c) \cdot \exp\left(-\frac{\Delta\hat{H}(\mathbf{x}_c)}{RT}\right) \tag{20}$$

where $\hat{K}_0(\mathbf{x}_c) = \text{softplus}(\text{NN}_{K_0}(\mathbf{x}_c)) > 0$ and $\Delta\hat{H}(\mathbf{x}_c) = -\text{softplus}(\text{NN}_H(\mathbf{x}_c)) < 0$ (exothermic adsorption, physically required for physisorption).

By construction, $\partial \ln \hat{K}/\partial(1/T) = -\Delta\hat{H}/R$ exactly. The predicted $\Delta\hat{H}$ values (in kJ/mol) are physically interpretable adsorption enthalpies, validatable against calorimetric measurements (-15 to -35 kJ/mol for methane-coal). $\mathcal{L}_{\text{vH}}$ is dropped.

2.6 Constraint Subsumption Map

A key structural insight emerging from the corrected formulations is that the Sips gauge architecture (Strategy 3) subsumes the bounds guarantee (Strategy 1) and the pressure-monotonicity guarantee (Strategy 2), as long as all Sips parameters are positive and raw $P > 0$ is used. This has direct implications for which ablation variants are independent.

Strategy	Bounds	Mono	Sips form	Van't Hoff
Strategy 1 (sigmoid head)	✓	—	—	—
Strategy 2 (NN^+ pathway)	—	✓	—	—
Strategy 3 (Sips gauge)	✓	✓	✓	—
Strategy 4 (Arrhenius)	—	—	—	✓

Strategies 3 and 4 together (Sips gauge + Arrhenius) enforce all four physical constraints and form the all-hard variant V2.

2.7 Revised Factorial Ablation Design

v3 revision: The ablation design is revised to correct two issues from v2: (a) the old V8 (Hard-{S+B+M}) was redundant because the Sips gauge already enforces bounds and monotonicity; (b) the Sips×vH interaction was confounded in the old V2/V8 because bounds and monotonicity were simultaneously hard. The new V8 (Hard-{S+vH}) enables a clean Sips×vH interaction test using {V1, V5, V6, V8}, analogous to the clean Bounds×Monotonicity test using {V1, V3, V4, V7}.

Design: Eight scientifically motivated variants preserving all main effects and the two primary interaction subspaces:

ID	Variant	Bounds	Mono	Sips	vH	Purpose
V1	All-soft	S	S	S	S	Baseline
V2	All-hard	H*	H*	H	H	Maximum enforcement
V3	Hard-B	H	S	S	S	Isolate bounds
V4	Hard-M	S	H	S	S	Isolate monotonicity
V5	Hard-S	H*	H*	H	S	Isolate Sips (subsumes B+M)
V6	Hard-vH	S	S	S	H	Isolate thermal
V7	Hard-{B+M}	H	H	S	S	Joint bounds+mono; B×M interaction
V8	Hard-{S+vH}	H*	H*	H	H	Sips+thermal; S×vH interaction

* Enforced implicitly by the Sips gauge architecture.
H = hard, S = soft.

Interaction tests enabled:

- Bounds × Monotonicity: {V1, V3, V4, V7} — clean 2 × 2
- Sips × Van't Hoff: {V1, V5, V6, V8} — clean 2 × 2

Note on V5: Because the Sips gauge subsumes bounds and monotonicity, V5 is not a pure single-constraint isolate. This should be explicitly acknowledged in the manuscript. V5 tests the effect of the Sips gauge architecture as a whole (which includes implicit B+M enforcement), not the Sips functional form constraint in isolation. This is an inherent consequence of the architectural approach, not a design flaw, and is scientifically informative in its own right.

3 Implementation Pipeline

Pipeline design principle: The evaluation harness is built *before* any model, not after. Every model variant plugs into the same harness automatically. This prevents retrofitting, ensures fair comparison, and eliminates the risk of adapting evaluation procedures to favour certain results.

Pipeline Overview

Step	Component	Produces	Used by
1	Data pipeline + 3-tier splits	Loaders, scalers	All steps
2	Evaluation harness	Stats framework	Steps 4–10
3	Soft baseline + λ_p sweep	V1 + sensitivity	Manuscript §3.1
4	Hard bounds (V3)	Architecture A	Harness
5	Hard monotonicity (V4)	Architecture B	Harness
6	Hard Sips — gauge (V5)	Architecture C	Harness
7	Hard vH — Arrhenius (V6)	Architecture D	Harness
8	Joint variants V2, V7, V8	Final 3 variants	Harness
9	Full ablation + interaction	All 8 variants	Manuscript §3.2–3.5
10	UQ calibration	Calibrated models	Manuscript §3.6
11	Interpretability	SHAP + param plots	Manuscript §3.7

4 Step-by-Step Implementation

4.1 Step 1: Data Pipeline with Three-Tier Evaluation

Rationale: Extrapolation performance under distribution shift is the primary scientific argument for hard constraints. Three evaluation tiers must be defined from the start so every model is evaluated identically. GroupKFold at the experiment level prevents sample-level leakage across all splits.

v3 addition: An explicit assertion that $P > 0$ for all samples is required, since the Sips gauge monotonicity guarantee depends on raw positive pressure. Pressure values at or below zero are physically meaningless in this context and would produce NaN gradients in $(\hat{K}P)^{\hat{n}}$.

What it achieves:

- Experiment-level GroupShuffleSplit (no leakage)
- Log target transform + robust scaling (fitted on train only)

- 12-feature engineering (same as original study)
- 5-fold GroupKFold for cross-validation
- Three evaluation tiers: T1 (interpolation), T2 (rank extrap), T3 (pressure extrap)
- Raw pressure tensor preserved (unscaled MPa) for Sips output layer
- Assert $P > 0$ for all samples before training

Requirements — Step 1:

Implement a complete data pipeline in PyTorch for methane coal sorption modeling.

INPUT CSV columns: experiment_id, coal_rank (anthracite/lv-bitu/ mv-bitu/hv-bitu, temperature_K, pressure_MPa, moisture_wt, ash_wt, volatile_wt, CH4_adsorption_m3t

SPLITS (implement all, no overlap):

- Main split: GroupShuffleSplit 80/20 by experiment_id, stratify by joint temperature-pressure bin within groups
- T1 (interpolation): standard 20% held-out test set
- T2 (rank extrapolation): ALL lignite experiments completely withheld from training. If lignite in test set, move to T2. T2 evaluated separately.
- T3 (pressure extrapolation): from T1, isolate samples with P > 8.0 MPa (top 10th percentile). Evaluate models on these without retraining.
- 5-fold CV: GroupKFold(n_splits=5) on training set, group=experiment_id

FEATURES: engineer Tr=T/190.6, Pr=P/4.60, FC=100-M-A-VM, OM=(VM+FC)/100, beta=1/(8.314*T), PT=P*T, MV=moisture*volatile. Total: 12 features. Robust scale (median/IQR) fitted on train fold only.

PRESSURE HANDLING (critical for Sips gauge):

- Return raw_pressure tensor (unscaled MPa) alongside scaled feature tensor. The Sips output layer uses raw pressure.
- assert (raw_pressure > 0).all(), "Sips monotonicity guarantee requires P > 0 for all samples"
- Log minimum pressure in dataset; warn if any P < 0.01 MPa

RETURN: DataLoaders (train/val/T1/T2/T3), scaler object, feature_names list, fold_indices for CV, raw_pressure tensor (unscaled MPa), experiment_id tensor.

VERIFICATION: assert zero experiment overlap across splits. Assert T2

contains only lignite. Assert T3 subset of T1. Assert P > 0. Print split sizes and rank distribution per split.

4.2 Step 2: Unified Evaluation Harness

Rationale: Building the evaluation harness before any model ensures all variants are compared on identical metrics computed identically. Statistical rigor (bootstrap CIs, corrected tests) is applied from the first result. Physical violation rates with uncertainty bounds are treated as first-class results.

Requirements — Step 2:

Implement a unified evaluation harness for PINN constraint comparison.

FUNCTION: evaluate_model(model, data_dict, n_bootstrap=100, n_mc=100, alpha=0.05)

ACCURACY METRICS (compute on T1, T2, T3 separately):

- R2, RMSE, MAE, MaxAE on original scale (back-transform)
- Median relative error (%), mean relative error (%)
- 95% bootstrap CI for each metric (stratified by coal rank)

PHYSICAL VIOLATION METRICS (primary):

- neg_rate: fraction samples with q_hat < 0 (+ 95% CI)
- sat_rate: fraction with q_hat > q_max (+ 95% CI)
- nonmono_rate: fraction with dq/dP < 0 at any P in linspace(0.01, 12, 50) per sample (+ 95% CI)
- vH_mae: mean |d(lnK)/d(1/T) + dH/R| (+ 95% CI)
- For hard constraints: assert 0.0 and raise ValueError if non-zero (structural guarantee must hold)

UQ METRICS (MC Dropout, 100 passes):

- sigma_epi, sigma_ale, sigma_total
- ECE (4 confidence levels: 68, 90, 95, 99%)
- rho_s: Spearman(|error|, sigma_total)
- NLL, CRPS, Sharpness (95% PI width)
- Coverage at 68%, 90%, 95%, 99%

COMPUTATIONAL METRICS:

- train_time_sec, inference_time_ms_per_sample

- n_parameters, memory_MB (peak GPU)
- epochs_to_R2_085, epochs_to_R2_090

STATISTICAL COMPARISON FUNCTIONS:

- compare_variants(v1, v2, metric): bootstrap paired t-test + Cohen's d + p-value
- bonferroni_correct(p_values, n=28): corrected alpha = 0.05/28 = 0.0018
- anova_2x2(results, factor1, factor2, metric)
- mcnemar_test(violations_v1, violations_v2)

OUTPUT: dict with all metrics + CIs, auto-generate LaTeX table. Log everything to CSV for reproducibility.

4.3 Step 3: Soft Baseline + λ_p Sensitivity Analysis

Rationale: Before declaring hard constraints superior, we must rule out that soft baseline underperforms simply due to poor λ_p tuning. The sensitivity sweep is mandatory to make the hard vs. soft comparison credible.

Requirements — Step 3:

Implement the soft-constraint PINN baseline (V1) and lambda_p sensitivity sweep.

ARCHITECTURE (V1 baseline):

- Input: 12 features (including scaled pressure)
- 5 residual blocks [512,1024,512,256,128], LayerNorm (NOT BatchNorm), Swish activations, dropout=0.1
- 3 output heads: mean (softplus for positivity --- soft), log-variance (aleatoric), physics params [q_max,K,n] via softplus
- All 4 physics losses active as soft penalties

LAMBDA_P SENSITIVITY SWEEP:

- Train V1 with lambda_p in [0.0, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0, 2.0]
- For each: 3 seeds, 5-fold CV
- Record: R2, RMSE, violation rates, ECE
- Select lambda_p* = argmin validation RMSE
- Plot: dual-axis RMSE vs lambda_p and violation rate vs lambda_p (Pareto trade-off)

- KEY RESULT: is there a lambda_p achieving zero violations AND high R2? (Expected: no)

TRAINING: AdamW, lr=1e-3, wd=1e-5, early stopping patience=100, max_epochs=1000. Log per epoch: loss, R2_val, violation_rates.
Plug into harness. Store in results_dict['V1'].

4.4 Step 4: Hard Bounds Variant (V3)

Rationale: Physical bounds are the simplest constraint to enforce exactly and have the smallest computational overhead. Even this trivial hard constraint establishing measurable benefit over soft enforcement anchors the argument for the full study.

What it achieves: V3 (Hard-B) with corrected sigmoid-only output head that structurally maps to $(0, \hat{q}_{\max})$. $\mathcal{L}_{\text{bounds}}$ removed; all other losses unchanged.

Requirements — Step 4:
Implement V3 (Hard-B): soft baseline with hard physical bounds. Modify ONLY the mean prediction head from V1:
class HardBoundsHead(nn.Module):

- head_qmax: Linear -> softplus -> q_max > 0
- head_q: Linear -> raw scalar z_q
- forward(h):
  - q_max = softplus(head_qmax(h))
  - z_q = head_q(h)
  - q_hat = q_max * torch.sigmoid(z_q)
  - # q_hat is strictly in (0, q_max) for all finite z_q
  - return q_hat, q_max

VERIFICATION:

- verify_bounds(model, loader): assert (q_hat > 0).all() and (q_hat < q_max).all()
- Run on 1000 random inputs including extreme P values
- Raise ValueError with violation details if assertion fails

Remove L_bounds. Keep L_Sips + L_mono + L_vH soft. Plug into harness. Store in results_dict['V3'].

4.5 Step 5: Hard Monotonicity via Non-Negative Pressure Pathway (V4)

Rationale: Monotonic pressure-adsorption is a thermodynamic law (Le Chatelier). Violations are not just statistically undesirable — they are physically impossible. Soft constraints can produce non-monotonic isotherms for rare compositional inputs at high pressure, exactly the extrapolation scenario most relevant for reservoir assessment.

The corrected architecture directly guarantees $\partial\hat{q}/\partial P \geq 0$ through non-negative weights and monotone activations on the pressure pathway, without relying on convexity arguments.

Requirements — Step 5:

Implement V4 (Hard-M): non-negative pressure pathway for hard monotonicity enforcement.

ARCHITECTURE:

- Compositional encoder: standard 5 residual blocks [512,1024,512,256,128] with Swish. Input: 11 compositional features (no pressure). Output: h_comp (128-dim)
- Pressure pathway: separate 3-layer network with hidden dims [64, 32, 16]. CRITICAL:
  - All weight matrices on pressure-input connections reparameterized: W = F.softplus(W_raw) >= 0
  - Activations: softplus ONLY (not Swish, not ReLU). Swish is non-monotone and breaks the guarantee.
  - Pressure P enters ONLY this pathway, nowhere else.
- Output: q_hat = linear(concat(h_comp, h_pressure))

REPARAMETERIZATION (critical --- must be in forward pass):

- def forward(self, x_comp, P_raw):
- W_eff = F.softplus(self.W_raw) # recompute each pass
- h_P = softplus(W_eff @ P_raw + self.b)
- Do NOT store softplus(W_raw) as a parameter.

VERIFICATION:

- verify_monotonicity(model, n_comp=500, n_P=50):
- Sample 500 random coal compositions from training dist

- Sweep P in linspace(0.01, 15, 50)
- dq_dP = torch.autograd.grad(q_hat.sum(), P_raw)[0]
- assert (dq_dP >= -1e-6).all()
- Report: max violation magnitude, fraction negative

Remove L_mono. Keep L_Sips + L_bounds + L_vH soft. Plug into harness. Store in results_dict['V4'].

4.6 Step 6: Gauge Function Sips Architecture (V5)

Rationale: The Sips gauge variant enforces the functional form of the pressure-adsorption relationship and, as a structural consequence, also guarantees bounds and monotonicity. The encoder predicts physically interpretable parameters $[\hat{q}_{\max}, \hat{K}, \hat{n}]$ from composition alone, enabling external validation against classical isotherm literature and SHAP attribution per parameter.

The manuscript should explicitly acknowledge that V5 is not a single-constraint isolate but a composite architectural choice. The scientific value of V5 is both in testing the Sips functional form constraint and in demonstrating the interpretability bonus of the gauge approach.

Requirements — Step 6:

Implement V5 (Hard-S): gauge function Sips PINN.
class SipsGaugePINN(nn.Module):

- Shared encoder: 5 residual blocks [512,1024,512,256,128], input=11 compositional features (NO pressure in encoder)
- head_qmax: Linear(128,1) -> softplus -> q_max > 0
- head_K: Linear(128,1) -> softplus -> K > 0
- head_n: Linear(128,1) -> softplus -> n > 0
- head_logvar: Linear(128,1) -> log_sigma2
- forward(x_comp, P_raw):
  - assert (P_raw > 0).all(), "Sips requires P > 0"
  - h = encoder(x_comp)
  - q_max, K, n = heads(h)
  - KP_n = (K * P_raw).pow(n)
  - q_hat = q_max * KP_n / (1 + KP_n)
  - return q_hat, log_sigma2, q_max, K, n

STRUCTURAL VERIFICATION:

- Assert q_hat > 0 and q_hat < q_max for all samples
- Assert dq/dP >= 0 for linspace(0.01, 15, 50)
- These should hold by construction; any violation indicates a bug (not a model limitation)

PARAMETER ANALYSIS:

- Compute mean q_max, K, n per coal rank
- Assert q_max in [10,40] m3/t, K in [0.1,5] MPa-1, n in [0.3,2.0]
- SHAP on each parameter head separately

Remove L_Sips, L_bounds, L_mono (all subsumed). Keep L_vH soft. Plug into harness. Store in results_dict['V5'].

4.7 Step 7: Arrhenius Reparameterization (V6)

Rationale: Van't Hoff consistency governs temperature dependence — critical for predictions at temperatures outside the narrow 20–50°C training range. The Arrhenius reparameterization produces interpretable $\Delta\hat{H}$ values validatable against calorimetric literature.

Requirements — Step 7:

Implement V6 (Hard-vH): Arrhenius K(T) reparameterization. Modify the physics parameter head from V1:

class ArrheniusHead(nn.Module):

- head_K0: Linear(128,1) -> softplus -> K0 > 0
- head_dH: Linear(128,1) -> -softplus(x) -> dH < 0 (exothermic)
- forward(h, T_raw): K = K0 * torch.exp(dH / (8.314 * T_raw)) (note: dH < 0, so exponent is negative as required)
- Return K, K0, dH (convert to kJ/mol for display)

VERIFICATION:

- Compute d(lnK)/d(1/T) numerically for 100 samples
- Assert max|d(lnK)/d(1/T) + dH/R| < 1e-4
- Report dH distribution per coal rank in kJ/mol
- Assert 95% of dH in [-60, -5] kJ/mol

Compare dH to Busch & Gensterblum (2011): anthracite -22, bituminous -18, lignite -14 kJ/mol. Plug into harness. Store in results_dict['V6'].

4.8 Step 8: Joint Variants V2, V7, V8

Rationale: Joint variants test whether hard constraints combine additively or otherwise. V7 (Hard-{B+M}) is the most practically relevant: zero-violation guarantee on the two most commonly violated constraints at minimal cost. V8 (Hard-{S+vH}) enables the clean Sips×vH interaction test. V2 (all-hard) tests whether maximum enforcement hurts accuracy.

Requirements — Step 8:

V7 (Hard-B+M): combine HardBoundsHead (Step 4) with non-negative pressure pathway (Step 5). Drop L_bounds and L_mono.

V8 (Hard-S+vH): SipsGaugePINN (Step 6) with ArrheniusHead (Step 7). All four physical losses dropped. L_data only. Note: Sips gauge already enforces bounds and mono implicitly.

V2 (All-hard): identical to V8. Sips + Arrhenius together enforce all four constraints. L_data only.

For each variant: run all structural verification functions. If V2 shows degraded accuracy vs V1, report this honestly. Store in results_dict.

NOTE: V2 and V8 are equivalent under this ablation design. Run as one variant. Report in the manuscript that the all-hard model is achieved by combining Sips gauge + Arrhenius, which is also V8.

4.9 Step 9: Full Ablation + Interaction Analysis

Requirements — Step 9:

MAIN EFFECTS TABLE: for all 8 variants on T1, T2, T3:

- R2 (mean +/- 95% CI), RMSE, MAE, MaxAE
- violation rates with CIs (hard variants must show 0.0)
- ECE, rho_s, NLL, train time, parameters, memory

PAIRWISE STATISTICAL TESTS (28 pairs):

- Bootstrap paired t-test on RMSE (100 bootstrap samples)
- Bonferroni correction: alpha = 0.05/28 = 0.0018
- Cohen's d for effect size
- McNemar's test for violation rate proportions
- Report as significance matrix (heatmap)

ANOVA INTERACTION ANALYSIS:

- Bounds x Mono: use V1(S,S), V3(H,S), V4(S,H), V7(H,H)
- Sips x vH: use V1(S,S), V5(H,S), V6(S,H), V8(H,H)
- Interaction delta: (V7-V1) - (V3-V1) - (V4-V1) and (V8-V1) - (V5-V1) - (V6-V1)
- delta > 0: superadditive; delta < 0: antagonistic

EXTRAPOLATION ADVANTAGE:

- R2 retention ratio: R2_T2/R2_T1 and R2_T3/R2_T1 per variant
- Plot: retention ratio vs violation rate (expected negative correlation)

4.10 Step 10: Calibrated UQ Across All Variants

Rationale: A key hypothesis is that hard constraints, by narrowing the admissible solution space, produce lower epistemic uncertainty. If confirmed, this provides a mechanistic explanation for ensemble collapse observed in the original paper, connecting the two studies theoretically.

Requirements — Step 10:

For each variant (100 MC Dropout passes):

- Compute epistemic_fraction% per variant
- Fit tau via ECE minimization on validation set
- Evaluate calibrated ECE, rho_s, NLL, CRPS on T1, T2, T3

KEY PLOT: epistemic_fraction vs constraint_hardness_score (score = number of hard constraints active, 0--4). Expected: monotone decreasing relationship.

CALIBRATION UNDER EXTRAPOLATION:

- ECE on T1 and T2 separately per variant
- Report ECE_T2/ECE_T1 ratio (hard constraints expected to maintain better calibration under shift)

4.11 Step 11: Interpretability and Physical Parameter Validation

Rationale: The gauge function variant (V5) produces parameters validatable against external literature — impossible for standard PINNs. SHAP per parameter head reveals what compositional features control maximum capacity, equilibrium constant,

and heterogeneity. This is new scientific knowledge about coal sorption, not just model evaluation.

Requirements — Step 11:

GAUGE FUNCTION PARAMETER ANALYSIS (V5):

- q_max, K, n distributions per coal rank (violin plots)
- Assert Pearson corr(q_max, volatile) < -0.3
- SHAP on head_qmax, head_K, head_n separately
- Compare to classical Langmuir fits from Step 3

ARRHENIUS dH VALIDATION (V6):

- dH mean +/- std per coal rank in kJ/mol
- Compare to Busch & Gensterblum (2011)
- One-way ANOVA + Tukey HSD across ranks

GLOBAL SHAP COMPARISON (V1 vs V5):

- Mean |SHAP| feature importance ranking for both
- Rank correlation between V1 and V5 importance

5 Key Connections Between Steps

Step produces	Used in	How
Step 1: T1/T2/T3 splits	All steps	Identical evaluation on all tiers
Step 1: raw_pressure	Steps 6,7,8	Required by Sips gauge output layer
Step 2: Harness	Steps 3–11	evaluate_model() for every variant
Step 3: λ_p^*	Steps 4–8	All soft components use optimal λ_p^*
Step 3: V1 results	Step 9	Baseline for all pairwise comparisons
Step 5: NN^+ pathway	Step 8 (V7)	Reused in joint variant
Step 6: Gauge code	Step 8 (V2/V8)	Reused in joint variants
Step 7: Arrhenius	Step 8 (V2/V8)	Combined with gauge for all-hard
Step 9: Ablation	Manuscript §3.2–3.5	Main results section
Step 10: UQ	Manuscript §3.6	Connects to original paper's ensemble finding
Step 11: Interp	Manuscript §3.7	New scientific knowledge contribution

6 What to Remember and Avoid

Critical design decisions to remember throughout:

1. **LayerNorm everywhere, not BatchNorm.** Hard constraint variants with small batches may expose BatchNorm instability. LayerNorm is batch-size independent.
2. **Softplus reparameterization in forward pass, not stored.** Non-negative weights for the pressure pathway must be recomputed at each forward pass via `F.softplus(W_raw)`. Do not store `softplus(W)` as a parameter.
3. **Softplus or sigmoid only on the pressure pathway.** Swish activations are globally used in compositional encoders but must not be used on the pressure pathway in V4/V7, as Swish is non-monotone near $x \approx -1.3$. Use softplus on all pressure-dependent connections.
4. **Pressure must not enter any unconstrained pathway (V4, V7).** If pres-

sure P is passed through any standard linear layer with unrestricted weights, the monotonicity guarantee is lost entirely. Verify at implementation time that the only pathway from P to the output is the reparameterized non-negative pathway.

5. **Assert $P > 0$ before every Sips gauge computation.** The Sips monotonicity guarantee requires physical pressure to be strictly positive. NaN gradients will occur at $P = 0$.
6. **V5 is not a single-constraint isolate.** The Sips gauge enforces bounds and monotonicity implicitly. Acknowledge this explicitly in the manuscript when interpreting V5's main effect. This is not a weakness — it is the architectural consequence of choosing a physically structured output layer.
7. **Identical seeds and hyperparameters across variants.** Any difference in initialization or training hyperparameters confounds the comparison. Only the constraint enforcement strategy changes between variants.
8. **Report negative results honestly.** If all-hard (V2) underperforms all-soft (V1) on interpolation accuracy, this is a publishable finding. The paper's value is the rigorous comparison.
9. **Bonferroni correction is mandatory.** With 28 pairwise comparisons, uncorrected p-values at $\alpha = 0.05$ produce approximately 1.4 false positives by chance. Corrected $\alpha = 0.0018$.
10. **Extrapolation results are the primary contribution.** If hard and soft constraints perform similarly on interpolation but hard constraints dominate on T2/T3, the abstract and conclusions lead with this finding.

7 Project Report Structure

7.1 Tentative Title

"Guaranteeing Physical Validity by Construction: A Systematic Factorial Evaluation of Hard vs. Soft Thermodynamic Constraint Enforcement in Physics-Informed Neural Networks for Sorption Studies"

7.2 Report Structure

1. Introduction

- Physical validity in safety-critical ML: why violations matter
- Soft constraint limitations: tuning-dependent, violation-tolerant, extrapolation-fragile
- Hard constraint PINNs: state-of-art in PDEs, gap in sorption

- Three architectural strategies and their subsumption relationships
- Three research questions and six contributions

2. Materials and Methods

- Dataset: 993 measurements, 114 experiments (reference original)
- Three-tier evaluation framework (T1/T2/T3) with rationale
- Soft constraint PINN baseline: architecture + λ_p formulation
- Hard bound enforcement: sigmoid-bounded output activation
- Hard monotonicity: non-negative pressure pathway with monotone activations
- Hard Sips: gauge function parameterization; subsumption of bounds and mono
- Hard van't Hoff: Arrhenius structural reparameterization
- Revised factorial ablation: 8 variants, two clean 2×2 interaction subspaces
- Unified evaluation harness: all metrics, bootstrap CIs, McNemar, ANOVA
- UQ: MC Dropout + temperature scaling for all variants
- Interpretability: SHAP per parameter head, ALE, physical validation

3. Results and Discussion

- **3.1** λ_p sensitivity: soft constraint performance ceiling with optimal tuning
- **3.2** Main effects: interpolation comparison (T1), 8 variants
- **3.3** Physical violation rates: hard constraints achieve zero; statistical comparison
- **3.4** Extrapolation analysis (T2, T3): hard constraint advantage under distribution shift
- **3.5** Constraint interaction effects: ANOVA results on two clean 2×2 subspaces; superadditive/antagonistic findings
- **3.6** Uncertainty calibration: epistemic fraction vs. constraint hardness; calibration under extrapolation
- **3.7** Interpretability: gauge parameter distributions, $\Delta\hat{H}$ validation, SHAP per parameter head
- **3.8** Computational cost analysis: Pareto frontier (accuracy / violation rate / training time)
- **3.9** Negative results and limitations (honest reporting)

4. Conclusions

5. Repository [Closed initially; structured with scripts, run-files, reports, checkpoints, figures, and metrics, with a readme. To be considered for open access after review.]